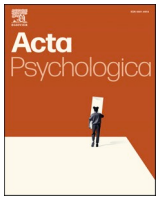
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Problematic social media use among French psychology university students: Prevalence, correlates, and mediation–moderation analyses

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ABSTRACT

Background: Problematic social media use is rising among university students, harming mental health, academic performance, and self-regulation; psychology students are especially relevant due to addiction training yet high social media exposure.
Objectives: This study aimed to (1) estimate the prevalence of PSMU among French psychology students, (2) identify its sociodemographic, behavioral, social, psychological, mental health, and academic correlates and predictors, and (3) examine mediation and moderation mechanisms linking social media use frequency (SMUF) to PSMU.
Methods: A cross-sectional online survey was conducted with 973 university students. PSMU was measured using the Bergen Social Media Addiction Scale (BSMAS). Data analyses included correlational analyses, multivariate linear regression, and mediation and moderation models.
Results: The mean PSMU score (M = 2.22, SD = 0.76) was below the scale midpoint. Overall, 28.9% of participants scored above the scale midpoint (>2.5). Using progressively more stringent criteria, the estimated prevalence was 15.0% based on the descriptive threshold ("descriptive prevalence"), 5.6% based on the high-risk threshold ("risk prevalence"), and 3.8% based on meaningful functional impairment ("clinical prevalence"). Strong predictors of PSMU included social media-related problems, perceived need to control use, SMUF, impulsivity (negative urgency), and depressive mood. Women and sexual minority students reported higher PSMU than men and heterosexual, respectively; although some of the effect-sizes for differences across sexual orientation categories were relatively small. Social resources (e.g., relationship satisfaction, perceived support) were negatively associated with PSMU but did not moderate the SMUF–PSMU relationship. Sleep difficulties, depressive mood, and self-esteem partially mediated the association between SMUF and PSMU.
Conclusions: The study suggests that the PSMU among psychology students might be mainly driven by emotional vulnerability and self-regulation deficits rather than usage frequency alone. Interventions should prioritize emotional regulation, sleep quality, and mental health over simply reducing screen time.

1. Background

Problematic social media use (PSMU) refers to a pattern of excessive, compulsive, and poorly controlled engagement with social media platforms that is associated with psychological distress and/or functional impairment in daily life (Andreassen et al., 2016; Andreassen & Pallesen, 2014; Montag et al., 2021; Moretta & Wegmann, 2025). Although PSMU shares several characteristics with behavioral addictions—such as salience, impaired control, conflict, and relapse—it is best conceptualized as a maladaptive pattern of use rather than a formally recognized addictive disorder (Moretta & Wegmann, 2025). Unlike gambling disorder, which is included in major diagnostic classification systems, PSMU has not been established as a distinct psychiatric diagnosis in the Diagnostic and Statistical Manual of Mental Disorders (DSM-5-TR) (American Psychiatric Association (APA), 2022) or the International Classification of Diseases (ICD-11) (World Health

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Organization (WHO), 2022). Consequently, PSMU is generally viewed as a dimensional construct that captures problematic engagement with social media, beyond high-frequency use alone, and is thought to arise from interactions among reward sensitivity, social reinforcement, and deficits in self-regulation (Moretta & Wegmann, 2025).

1.1. Prevalence and correlates of problematic social media use

Epidemiological studies indicate that PSMU is common among adolescents and young adults, particularly university students, who represent one of the most intensive user groups of social media platforms (Andreassen et al., 2016; Carbonell-Colomer et al., 2025; Cheng et al., 2021). However, reported prevalence rates vary considerably, ranging from approximately 5% to over 25% in student populations (Bányai et al., 2017; Cheng et al., 2021). This wide variability reflects important methodological and conceptual differences across studies. First, researchers use different assessment instruments, which are based on distinct theoretical models and operationalize PSMU using different symptom sets and scoring procedures (Carbonell-Colomer et al., 2025; Cheng et al., 2021). Second, prevalence estimates are highly sensitive to the choice of cut-off criteria, with some studies applying conservative thresholds intended to identify individuals with marked functional impairment, whereas others use more liberal criteria that capture milder or subclinical patterns of problematic use (Bányai et al., 2017; Carbonell-Colomer et al., 2025). Third, differences in sample characteristics—including age, sex distribution, academic discipline, and cultural context—as well as the period of data collection may further contribute to inconsistent prevalence estimates. For example, social media engagement increased substantially during the COVID-19 pandemic, potentially inflating estimates reported during that period compared with pre-pandemic studies (Cheng et al., 2021). European studies generally report prevalence rates between 8% and 18% among university students, with higher estimates often observed in samples of social science and psychology students (Casale et al., 2023; Kuss & Griffiths, 2017; Shannon et al., 2022). Collectively, these findings highlight the absence of universally accepted diagnostic criteria for PSMU and emphasize the need for country-specific epidemiological studies using validated instruments and transparent, empirically justified cut-off thresholds to facilitate meaningful comparisons across populations.

Sociodemographic characteristics have been consistently associated with PSMU. Younger age is one of the most robust predictors, reflecting developmental vulnerabilities in impulse control and identity formation (Boer et al., 2021; Casale et al., 2023; Shannon et al., 2022). Female students generally report higher levels of PSMU than their male counterparts (Andreassen et al., 2016; Casale et al., 2023; Shannon et al., 2022). Although this difference has often been attributed to greater engagement in social comparison and appearance-related content, accumulating evidence suggests that it is likely driven by multiple interacting psychological and behavioral factors. Compared with men, women tend to use social media more frequently for interpersonal communication, relationship maintenance, emotional expression, and social support, which may increase emotional investment in online interactions (Alwuqaysi, 2025; Cheng et al., 2021; Kuss & Griffiths, 2017). Women also report higher levels of fear of missing out (FoMO), stronger sensitivity to social feedback, and greater use of social media for emotion regulation and coping with stress or negative affect, all of which have been identified as risk factors for PSMU (Fioravanti et al., 2021). Furthermore, greater exposure to appearance-related content and cyber-social comparison may reinforce repetitive checking behaviors and psychological dependence on social media platforms through self-reinforcing comparison processes (Arigo et al., 2024; Rousseau & Rodgers, 2025). These findings suggest that gender differences in PSMU reflect a complex interplay of emotional, interpersonal, and sociocultural processes rather than social comparison alone. Sexual minority individuals (e.g., bisexual and homosexual students) appear to be at elevated risk, possibly because social media provide alternative spaces for identity exploration and social support amid offline marginalization (Fatima et al., 2025; Pazdur et al., 2025; Valkenburg et al., 2022). Lower income, undergraduate status, and single relationship status have also been linked to higher PSMU, reflecting both structural stressors and greater reliance on online social interaction (Fatima et al., 2025; Fioravanti et al., 2025; Pazdur et al., 2025).

Behavioral patterns of social media use represent central proximal determinants of PSMU. High frequency of use for non-essential purposes, experiences of academic or professional problems due to social media, and a perceived need to control usage are consistently associated with higher PSMU scores (Billieux et al., 2015; Elhai et al., 2017). Although the use of smartphone applications designed to monitor or restrict social media use reflects awareness of problematic behavior, there are some evidence that such tools often fail to reduce PSMU in the absence of broader self-regulatory capacities (Aboujaoude et al., 2022; Fioravanti et al., 2025; ven Deursen et al., 2015; Vera Cruz et al., 2023). These findings suggest that frequency of use alone is insufficient to explain PSMU without considering subjective loss of control.

PSMU has been consistently associated with psychological factors, particularly deficits in executive functioning and personality traits. Difficulties in working memory, inhibitory control, and attentional regulation increase vulnerability to habitual and compulsive social media checking by reducing individuals' ability to resist immediate online rewards and maintain focus on long-term goals (Fioravanti et al., 2025; Wilmer et al., 2017). These self-regulatory processes are especially relevant for university students, including psychology students, whose academic success depends heavily on sustained attention, critical reading, information integration, and effective self-directed learning. Frequent interruptions from social media may therefore have particularly detrimental effects on this population by undermining concentration, learning efficiency, and academic performance. Likewise, impulsivity—especially urgency and lack of perseverance—has emerged as a robust risk factor across problematic use of digital technology, including PSMU, as it predisposes individuals to engage in automatic and poorly controlled online behaviors despite negative consequences (Billieux et al., 2021). Personality research further indicates that high neuroticism and low conscientiousness are consistently associated with greater PSMU, reflecting increased emotional vulnerability and reduced self-discipline, respectively (Casale & Banchi, 2020; Đurić et al., 2024; Kircaburun et al., 2020). Conversely, a stronger sense of future self-continuity, which promotes self-regulation, delayed gratification, and goal-directed behavior, has been associated with lower levels of problematic digital media use (Sokol & Serper, 2020). Together, these findings suggest that individual differences in executive functioning and self-regulation may be particularly important for understanding PSMU among psychology students, whose academic training places substantial demands on these cognitive capacities (Ma, 2025; Pang et al., 2025; Vera Cruz et al., 2026; Vera Cruz & Da Rosa, 2026).

A substantial body of research links PSMU to adverse mental health outcomes, although the available evidence remains largely correlational and does not establish definitive causal relationships (Ahmed et al., 2024; Corke et al., 2025; Karim et al., 2020). Sleep difficulties, including delayed sleep onset, poor sleep quality, and nocturnal awakenings, are frequently reported among individuals with high PSMU, partly due to bedtime social media use, prolonged screen exposure, and heightened cognitive and emotional arousal before sleep (Alimoradi et al., 2019). PSMU has also been consistently associated with depressive symptoms and negative affect (Ahmed et al., 2024; Corke et al., 2025). Longitudinal studies suggest that this relationship is likely bidirectional: elevated PSMU may contribute to subsequent depressive symptoms through mechanisms such as excessive social comparison, cybervictimization, disrupted sleep, and reduced engagement in offline activities, whereas individuals experiencing depressive symptoms may increasingly rely on social media as a means of mood regulation,

distraction, social reassurance, or escape from negative emotions, thereby increasing their risk of developing problematic patterns of use (Bányai et al., 2017; Marino, 2018; Valkenburg et al., 2022). Similarly, lower self-esteem has been identified as both a predictor and a consequence of PSMU, with reciprocal associations explained by increased dependence on online validation and repeated upward social comparison (Corke et al., 2025; Vogel et al., 2014). Collectively, these findings support conceptual models in which psychological well-being and PSMU influence one another over time and suggest that mental health variables may mediate or reinforce the relationship between social media use frequency and the severity of problematic use.

Among university students, PSMU has been associated with poorer academic functioning (Ma, 2025; Pang et al., 2025). Smartphone-related multitasking and frequent interruptions during study time predict reduced concentration, lower academic engagement, and diminished academic performance (Bowman et al., 2010; Lepp et al., 2015; Rosen, 2017). While objective academic grades tend to show small-to-moderate negative associations with PSMU, the perceived interference of social media with academic work appears to be a stronger and more consistent correlate (Junco, 2012). These findings suggest that PSMU primarily influences academic outcomes through cognitive mechanisms, including fear of missing out (FoMO)-driven hypervigilance, attentional overload, memory deficiencies, and impaired self-regulatory capacities (Ma, 2025; Pang et al., 2025).

1.2. The present study

Despite the growing international literature, data on the prevalence and determinants of PSMU among French university students remain scarce, particularly within psychology programs. The present study therefore aimed to (1) estimate the prevalence of PSMU among French psychology students; (2) examine the correlational and predictive roles of sociodemographic, behavioral, social, psychological, mental health, and academic variables; and (3) test moderation and mediation models in which social variables moderate, and mental health variables mediate, the relationship between social media use frequency and PSMU. Addressing these objectives will contribute to a more comprehensive understanding of the vulnerability and protective factors associated with PSMU in an academically relevant population.

Psychology students constitute a particularly informative population for investigating PSMU for both empirical and theoretical reasons. University students are among the most intensive users of smartphone and social media and consistently exhibit higher rates of PSMU than the general adult population (Carbonell-Colomer et al., 2025; Casale et al., 2023; Cheng et al., 2021; Vera Cruz et al., 2026). Within higher education, psychology programs are characterized by a predominance of female students, a demographic group that generally reports higher levels of PSMU than males (Arigo et al., 2024; Rousseau & Rodgers, 2025; Shannon et al., 2022). Moreover, several studies have shown that psychology students experience relatively high levels of academic stress, anxiety, depressive symptoms, and emotional exhaustion compared with students in many other disciplines, factors that have been consistently associated with greater vulnerability to problematic social media use (Beiter et al., 2015; Casale et al., 2023; Ibrahim et al., 2013; Vera Cruz et al., 2026). At the same time, psychology students in France are exposed through their university curricula to coursework addressing mental health, addictive mechanisms and behaviors, and self-regulation. Such academic exposure may theoretically foster greater awareness of the psychological processes and potential risks associated with excessive social media use. However, the presence of addiction-related content in the curriculum does not, in itself, demonstrate that psychology students possess greater knowledge of these mechanisms or are better protected against problematic social media use (PSMU). Rather, this curricular context makes psychology students a theoretically relevant population for examining the emotional, social, and self-regulatory factors associated with problematic social media engagement. Studying this population may therefore provide valuable insights into the complex interplay of risk and protective factors underlying PSMU, without presuming that academic exposure necessarily translates into greater awareness or reduced vulnerability.

1.3. Research questions and hypotheses

Four research questions (RQs) guided the present study. RQ1: what is the level of problematic social media use (PSMU) among participants? RQ2: to what extent sociodemographic, social media use behavior, social, psychological, mental health, and academic variables are associated with participants' PSMU scores? RQ3: what are the predictive value of these sociodemographic, behavioral, social, psychological, mental health, and academic variables in explaining variance in PSMU scores? RQ4: to what extent selected social, psychological, and mental health variables mediated or moderated the relationship between social media use frequency (SMUF) and PSMU.

The sociodemographic variables included age, gender, sexual orientation, relationship status, education level, and income. Based on prior empirical findings, we hypothesized that younger (vs. older) students, female (vs. male) students, bisexual or homosexual (vs. heterosexual) students, single (vs. partnered) students, undergraduate (vs. master's level) students, and students with relatively lower (vs. higher) income would report significantly higher levels of PSMU.

Social media use behavior variables comprised SMUF for non-essential activities, self-reported experiences of personal, professional, or academic problems attributable to social media use, and a perceived need to control SMUF. We hypothesized that higher levels of each of these behavioral indicators would be associated with a greater likelihood of reporting higher PSMU scores.

Social variables included (a) the frequency of leisure or entertainment activities conducted with others; (b) the number of close relationships (family members and friends) with whom participants engaged in leisure activities; (c) satisfaction with family and friendship relationships; (d) satisfaction with intimate relationships; (e) perceived social support from family and friends in times of need; (f) level of community engagement (e.g., involvement in nonprofit or community-oriented organizations); and (g) level of religious engagement. We hypothesized that higher levels of these social resources would be associated with lower levels of PSMU.

Psychological variables encompassed cognitive functioning (i.e., self-reported difficulties related to working memory), impulsivity, personality traits, and the degree of connection with one's future-self. We hypothesized that greater cognitive difficulties and higher impulsivity would be strongly associated with higher PSMU scores. In contrast, higher levels of conscientiousness were expected to be associated with lower PSMU, whereas higher levels of neuroticism were expected to be associated with higher PSMU. Furthermore, stronger connection with the future-self was hypothesized to be associated with lower levels of PSMU.

Mental health variables included sleep difficulties (i.e., difficulty initiating sleep and/or frequent nocturnal awakenings), depressive mood, and self-esteem. We hypothesized that greater sleep difficulties, higher depressive mood, and lower self-esteem would be associated with higher PSMU scores.

Academic variables included smartphone-related interference with university work (i.e., multitasking behaviors) and mean academic grade from the previous semester. We hypothesized that greater smartphone-related interference would be significantly associated with higher PSMU scores, whereas higher academic performance would be associated with lower PSMU levels.

Finally, with respect to moderation and mediation processes, we hypothesized that social variables would significantly moderate the relationship between SMUF and PSMU, such that stronger social resources would attenuate this association. Additionally, we hypothesized that mental health variables would partially and significantly mediate

the relationship between SMUF and PSMU.

1.4. Theoretical ground for the variables selection, research questions, and hypotheses

This study is mainly exploratory and not theory driven. That is, the choice of variables, research questions, and hypotheses was mainly based on the findings of previous studies discussed in this introduction section.

However, a strong theoretical foundation for the present study is provided by integrating complementary models of problematic digital media use. The Interaction of Person–Affect–Cognition–Execution (I-PACE) model (Brand et al., 2025, 2019, 2016) serves as the overarching framework, proposing that Internet-based services problematic use arises from interactions among individual predispositions (e.g., age, gender, personality, impulsivity), affective and cognitive processes (e.g., depression, self-esteem, executive functioning), and reinforcement mechanisms that ultimately impair self-regulation. This model supports the examination of sociodemographic, behavioral, psychological, mental health, and academic predictors of PSMU. Compensatory Internet Use Theory (Kardefelt-Winther, 2014) further suggests that individuals may engage excessively with social media to regulate negative emotions or compensate for unmet offline needs, providing a rationale for investigating depressive mood, self-esteem, and the mediating role of mental health variables. Self-Regulation Theory (Baumeister & Heatherton, 1996; Carver & Scheier, 1998) and Temporal Self-Regulation Theory (Hall & Fong, 2007) explain how deficits in executive functioning, impulsivity, and future-self continuity increase vulnerability to excessive social media use and interfere with academic performance. Finally, Uses and Gratifications Theory (Katz et al., 1973; Rubin, 2009) and Social Compensation Theory (McKenna & Bargh, 2000; Valkenburg & Peter, 2027) posit that social media are used to satisfy interpersonal and emotional needs, implying that stronger offline social resources (e.g., social support, relationship satisfaction, community engagement) may buffer against PSMU. Together, these complementary theories provide a coherent framework for the study's research questions and hypotheses regarding the determinants, predictors, and mediation and moderation mechanisms underlying PSMU.

1.5. Contribution to the literature

The present study extends previous research in several important ways. First, rather than examining a limited set of correlates, it simultaneously investigates demographic, behavioral, cognitive, emotional, motivational, and social factors associated with PSMU within a comprehensive multivariable framework, enabling the identification of their independent associations. Second, it estimates PSMU prevalence using three complementary approaches that reflect distinct conceptualizations of problematic use, thereby facilitating a more nuanced interpretation of prevalence estimates. Third, by focusing on psychology students, who are expected to possess greater knowledge of addictive behaviors, the study provides a unique opportunity to examine whether such knowledge is associated with lower levels of PSMU. Together, these contributions provide a more integrated understanding of PSMU among university students and inform priorities for future research and prevention efforts.

2. Methods

2.1. Inclusion criteria

Participants were eligible for inclusion if they (a) were 18 years of age or older, (b) had been active social media users for at least the previous 12 months, (c) were enrolled as undergraduate or master's-level students in psychology, (d) were registered at and attending courses in a French university or an equivalent higher education institution, and (e) voluntarily agreed to participate by providing informed consent.

2.2. Participants

A total of 1002 students from two French universities completed the online questionnaire. Twenty-nine participants were excluded due to failure on attention-check items embedded in the survey (see Statistical analyses), resulting in a final sample of 973 participants. Participants ranged in age from 18 to 43 years, with a strong overrepresentation of younger students (18–21 years: n = 853, 87.66%; 22–25 years: n = 104, 10.69%; ≥26 years: n = 16, 1.64%). While the sample seems skewed toward younger participants, it reflects the age range of the student population in French universities, especially in humanities and social science faculties, were more than 90% of the undergraduation and the master's degree students are between 18 and 22 years-old (Ministère de l'Enseignement Supérieurs, de la Recherche et de l'Espace, 2026). Also, the sample comprised predominantly women (female: n = 660, 67.83%; male: n = 303, 31.14%; other: n = 10, 1.03%). This gender distribution reflects enrollment patterns in French humanities and social sciences, where women are substantially overrepresented (70.9% in 2023; Observatoire des Inégalités, 2025). Regarding academic level, 761 participants (78.21%) were undergraduate students and 212 (21.79%) were master's students. Full sociodemographic characteristics are presented in Table S1, in Supplementary Material.

The sample is likely representative of the target population, as 74.6% of eligible students at the two participating universities completed the survey.

2.3. Recruitment and procedure

The two participating universities were selected based on the availability of faculty members and research assistants who were willing to facilitate participant recruitment. Undergraduate students were recruited during lectures held in large lecture halls that were attended by virtually all students enrolled at the corresponding academic level. Master's students were recruited during seminar sessions, which were likewise attended by nearly all students within their respective year of study. This recruitment strategy was designed to maximize participation and enhance the representativeness of the sample within each academic level. A professor or trained research assistant presented the study objectives and informed students that participation was voluntary, anonymous, and without any academic or personal consequences in case of refusal. Students willing to participate accessed the online survey via a provided link, electronically signed the informed consent form, and completed the questionnaire.

2.4. Measures

2.4.1. Sociodemographic variables

Collected sociodemographic variables included age, gender (male, female, other), sexual orientation (heterosexual, bisexual, homosexual, other), relationship status (five categories), educational level (undergraduate, master's), and monthly income (four categories ranging from less than €500 to more than €1500).

2.4.2. Social media use variables

Social media use frequency (SMUF) for non-essential activities over the previous three months was assessed using a six-point scale (<1 h to >9 h/day). Participants also reported whether they had experienced personal, academic, or professional problems related to social media use during the previous three months and whether they perceived a need to control their social media use during the past 12 months (yes/no). Problematic social media use (PSMU) was measured using the French translation of the 6-item Bergen Social Media Addiction Scale (BSMAS; Andreassen & Pallesen, 2014; Andreassen et al., 2016), grounded in

Griffiths' (2005) six-component addiction model (salience, tolerance, mood modification, relapse, withdrawal, and conflict). Items were rated on a five-point Likert scale (1 = *Very rarely* to 5 = *Very often*). Internal consistency in the present sample was good (Cronbach alpha coefficient [α] = 0.77). Although the BSMAS has been validated in the French adult population (Brailovskaia & Margraf, 2024), to our knowledge, its psychometric properties have not been specifically validated among university students in France.

2.4.3. Social variables

Social variables included: (a) number of leisure or social outings with others, (b) number of close family members or friends involved in leisure activities, (c) satisfaction with family and friend relationships, (d) satisfaction with intimate relationships, (e) perceived social support from family and friends, (f) community engagement, and (g) religious engagement. Variables (c)–(e) were assessed using 5-point scales, whereas community and religious engagement were assessed using 7-point scales. Each construct was measured using a single item.

2.4.4. Psychological variables

Cognitive difficulties related to working memory were assessed using a French adaptation of eight items from the Working Memory Questionnaire (WMQ; Vallat-Azouvi et al., 2012), covering short-term storage, attention, and executive control. Items were rated on a 5-point scale (1 = *Never* to 5 = *Always*). Internal consistency values for the three dimensions were acceptable: α = 0.65, α = 0.64, and α = 0.72, respectively (overall α = 0.81). Impulsivity was measured using eight items from the UPPS-P Impulsive Behavior Scale (Billieux et al., 2012), corresponding to positive and negative urgency. Items were rated on a 4-point Likert scale (1 = *Strongly disagree* to 4 = *Strongly agree*), with good internal consistency (α = 0.79 and 0.77; overall α = 0.83). Personality traits were assessed using the French version of the 10-item Big Five Inventory (BFI; Rammstedt & John, 2007). Internal consistency varied across dimensions (α for extraversion = 0.80; α for agreeableness = 0.23 [correlation between the two items of this dimension: $r = 0.131$, $p < .001$], α for consciousness = 0.68; α for neuroticism = 0.84, and α for openness = 0.27 [correlation between the two items of this dimension: $r = 0.158$, $p < .001$]). The Agreeableness and Openness dimensions of the BFI-10 exhibited low internal consistency. However, because their Cronbach's alpha coefficients are comparable to those reported in previous studies using ultra-short personality measures, and because the two items comprising each dimension were positively and significantly correlated, these dimensions were retained in the statistical analyses. Nevertheless, their limited internal consistency is acknowledged as a study limitation and should be considered when interpreting the findings. Connection with the future-self (Hershfield, 2023) was assessed using a single item rated on an 11-point scale (1 = *Not at all connected* to 11 = *Very much connected*).

2.4.5. Mental health variables

Sleep difficulties were assessed using a single item from the French version of the Sleep Health Index (Vézina-Im et al., 2022). Items were rated on a 5-point Likert scale (1 = *Never* to 5 = *Very often*). Happiness and depressive mood were measured using the Short Depression–Happiness Scale (Joseph et al., 2004). Items were rated on a 4-point Likert scale (1 = *Never* to 4 = *Often*). The happiness items were reverse-coded such that higher scores indicated greater depressive mood. Internal consistency was acceptable, with α of 0.77 for the happiness subscale and 0.66 for the depression subscale; the overall scale demonstrated an α of 0.77. Self-esteem was assessed using the French version of the Single-Item Self-Esteem Scale (Robins et al., 2001). Items were rated on a 4-point Likert scale (1 = *Very true for me* to 4 = *Not at all true for me*).

2.4.6. Academic variables

Smartphone-related interference with academic activities was assessed using the Mobile Phone Interference in Life questionnaire (David et al., 2015), separately for in-class and out-of-class contexts (α = 0.91 for both). Mean academic grades from the previous semester were reported using six ordinal categories based on the French 0–20 grading system.

2.5. Ethics

Participants gave digital informed consent for participating in the study and completing the survey. Participation was voluntary and restricted to those aged ≥18. All data was anonymously collected. The study was conducted in accordance with Declaration of Helsinki — Ethical Principles for Medical Research Involving Human Subjects (World Medical Association (WMA), 2024). Ethical approval for the research project (Avis n°2026-95/Dossier 2026-01.01) was obtained from the Ethics Committee of the University of Picardie Jules Verne.

2.6. Statistical analyses

The mean PSMU score served as the dependent variable. All other variables were treated as independent variables. Data quality checks included attention-check items, missing data, and outlier detection. After exclusions, 973 participants were retained for analysis. The attention-check items were as follow: “If you are reading this item carefully, please select the response option labeled ‘the label of the modality’.”

RQ1 was addressed using descriptive statistics and ANOVA analyses. Specifically, we estimated the prevalence of PSMU using three complementary approaches. The first approach, referred to as “descriptive prevalence” (Rochat et al., 2025; Vera Cruz et al., 2026), consisted of calculating the mean PSMU score and determining the proportion of participants whose mean scores exceeded successive thresholds on the 1–5 scale. These thresholds included the scale midpoint (2.5; liberal threshold), the midpoint plus 0.5 points (3.0), and higher values. Following previous recommendations, the principal descriptive prevalence estimate was based on a threshold of >3.5 (conservative threshold), corresponding to one full point above the midpoint of the scale. The second approach estimated “high-risk prevalence” by applying a threshold of >3.4. This cut-off was derived from the statistical analyses conducted by Peng et al. (2023), who combined latent profile analysis with receiver operating characteristic (ROC)-based sensitivity analyses to establish an empirically supported threshold for addiction assessment instruments based on the six-component model of addiction, such as the Bergen Social Media Addiction Scale (BSMAS). The third approach estimated “excessive use impairing functioning” prevalence (Rochat et al., 2025). Participants were classified as meeting this criterion if they obtained a mean BSMAS score > 3.5 and reported experiencing functional impairment by endorsing either “often” (4) or “very often” (5) on the BSMAS conflict item: “You use social media so much that it has had a negative impact on your job/studies”. These three prevalence estimates reflect distinct conceptual perspectives. The descriptive prevalence provides an overview of the distribution of problematic social media use within the sample. The high-risk prevalence identifies individuals who are likely to be at elevated risk and who may benefit from preventive or early intervention strategies. Finally, the excessive use impairing functioning prevalence represents the most conservative estimate, identifying individuals whose problematic social media use is accompanied by clinically meaningful functional impairment and is therefore more appropriate for clinical screening purposes (Rochat et al., 2025).

RQ2 involved univariate correlation analyses (Pearson or point-biserial, depending on variable type).

RQ3 was examined using multivariate linear regression. Only variables that were significantly associated with PSMU in the univariate analyses were entered into the multivariable regression model. This strategy reduced the number of predictors from the initial 36 to 28, minimized the risk of model overfitting and unnecessary

multicollinearity, and facilitated the estimation of independent associations while maintaining an adequate predictor-to-sample-size ratio (Babyak, 2004; Dormann et al., 2013; Harrell Jr., 2015).

RQ4 was tested using mediation and moderation analyses with bootstrapping (5000 resamples). The choice of 5000 bootstrap samples was based on methodological recommendations indicating that this number provides a reliable approximation of sampling distributions for indirect and conditional effects in mediation and moderation analyses (; Efron & Tibshirani, 1993; Hayes, 2018; Preacher & Hayes, 2008). In addition, with the respect of the moderation analysis, it must be noted that a sensitivity power analysis conducted using *G*Power indicated that, with $\alpha = 0.05$, power = 0.80, a sample size of 973 participants, 12 predictors in the full model, and six tested interaction terms, the study was able to detect interaction effects as small as $f^2 \approx 0.017$. This effect-size is slightly below Cohen's criterion for a small effect ($f^2 = 0.02$), suggesting that the study was adequately powered to detect small-to-large moderation effects.

Descriptive analyses were conducted using IBM SPSS Statistics (version 31), whereas inferential analyses were performed using Jamovi (version 2.4).

3. Results

3.1. Level of PSMU among participants (RQ1)

Table S1, in Supplementary Material, presents the descriptive statistics for all study variables.

As shown in Table S1, the mean PSMU score was slightly below the midpoint of the scale (M = 2.22, SD = 0.76; range 1–5; midpoint = 2.5).

The first approach to estimating prevalence yielded the following distribution of PSMU severity. Overall, 28.9% of participants had a mean PSMU score above the scale midpoint (>2.5), 15.0% scored above 3.0, 5.5% scored above 3.5, 1.4% scored above 4.0, and 0.4% scored above 4.5. Observed PSMU mean scores ranged from 1.00 to 4.67.

Using the second approach, which identifies individuals at high risk of problematic social media use, 5.6% of participants exceeded the recommended threshold (mean score > 3.4).

The third approach, which estimates the prevalence of excessive social media use associated with meaningful functional impairment, indicated that 3.80% of participants met both criteria: a mean BSMAS score > 3.5 and endorsement of the conflict item at the level of "often" or "very often."

Bisexual participants reported significantly higher PSMU scores than heterosexual participants, $t(971) = -3.09$, $p = .006$, $d = -0.21$ (M = 2.27, SD = 0.76 vs. M = 2.11, SD = 0.74). No significant differences in PSMU were observed between undergraduate and master's students, $t(971) = 1.40$, $p = .163$, $d = 0.11$.

Regarding social media behaviors, mean social media use frequency (SMUF) was slightly below the scale midpoint (M = 2.73, SD = 0.98; 1–6 scale). Overall, 18.1% of participants reported experiencing personal, academic, or professional problems due to social media use in the previous three months, and 69.7% reported a perceived need to control their SMUF during the past year.

Finally, PSMU scores were negatively correlated with academic performance ($r = -0.093$, 95% CI [−0.155, −0.031], $p = .004$), indicating that higher PSMU was associated with lower grades. However, given the study's cross-sectional design, causal inferences cannot be drawn.

3.2. Correlates of PSMU (RQ2)

Table 1 presents the correlations between independent variables (IVs) and the dependent variable (DV; PSMU). Positive coefficients indicate that higher IV values are associated with higher PSMU scores, whereas negative coefficients indicate inverse associations.

The vast majority of IVs were significantly correlated with PSMU.

Table 1
Univariate relationships between the independent variables and the score of problematic social media use: correlations analysis results.

Variable/measures	*r*	95% CI lower bound	95% CI upper bound	*p*
Sociodemographic variables				
Age	−0.101	−0.163	−0.038	0.002
Gender	0.104	0.042	0.166	0.001
Sexual orientation	0.026	−0.037	0.088	0.426
Relationship status*	−0.140	−0.202	−0.078	<0.001
Education level (measured in terms of years of schooling)*	−0.045	−0.107	0.018	0.163
Income	−0.113	−0.174	−0.050	<0.001
Social media use behavior measures				
Smartphone use frequency for non-essential activities (in last 3 months; measured in hours)	0.433	0.381	0.483	<0.001
Having experienced personal, professional, or academic problems due to smartphone use (previous three months)	0.453	0.401	0.501	<0.001
Perceived need to control smartphone use frequency (previous 12 months)	0.318	0.260	0.373	<0.001
Social activities, relationships, and social engagement measures				
Number of leisure or entertainment outings with others (previous 30 days)*	0.053	−0.010	0.115	0.101
Number of close relationships (family members and friends) with whom participants engaged in leisure activities (previous 30 days)*	−0.025	−0.088	0.038	0.439
Satisfaction (level of) with family members and friends relationship	−0.191	−0.251	−0.129	<0.001
Satisfaction (level of) with intimate relationships	−0.208	−0.267	−0.147	<0.001
Perceived (level of) social support from family and friends when in need	−0.178	−0.239	−0.117	<0.001
Community (level of) engagement*	−0.034	−0.097	0.029	0.291
Religious (level of) engagement	0.086	0.023	0.148	0.007
Psychological measures				
WMQ memory total mean score	0.236	0.175	0.294	<0.001
WMQ attention total mean score	0.307	0.249	0.363	<0.001
WMQ executive control total mean score	0.280	0.221	0.337	<0.001
WMQ total mean score	0.326	0.268	0.381	<0.001
UPPS-P positive urgency total mean score	0.274	0.215	0.331	<0.001
UPPS-P negative urgency total mean score	0.312	0.254	0.367	<0.001
UPPS-P urgency total mean score	0.341	0.284	0.395	<0.001
BFI extraversion total score*	0.045	−0.018	0.107	0.165
BFI agreeableness total mean score*	−0.044	−0.106	0.019	0.172
BFI consciousness total mean score	−0.249	−0.307	−0.189	<0.001
BFI neuroticism total mean score	0.205	0.144	0.265	<0.001
BFI openness total mean score	−0.095	−0.157	−0.033	0.003
Connection (level of) with future-self 20 years from the present	−0.125	−0.186	−0.062	<0.001
Mental health measures				
Sleep duration (from falling asleep until finally waking up; in last 6 months; self-estimation measured in hours)*	−0.037	−0.099	0.026	0.254

(*continued on next page*)

Table 1 (*continued*)

Variable/measures	*r*	95% CI lower bound	95% CI upper bound	*p*
Sleep difficulties (difficulty falling asleep and/or frequent waking up during the night; experienced in last 6 months)	0.199	0.138	0.259	<0.001
SDHS total mean score	0.298	0.240	0.354	<0.001
SISE self-esteem	−0.185	−0.245	−0.124	<0.001
Academic measures				
MPIL (outside classroom) total mean score	0.239	0.179	0.297	<0.001
MPIL (inside classroom) total mean score	0.179	0.117	0.239	<0.001
Grade mean for the last university semester	−0.093	−0.155	−0.031	0.004

r = correlation coefficient; CI = confidence interval; *p* = *p*-value (significance at $p < .05$).

* Variables non-significantly related to the participants' score of problematic social media use.

Only eight of the 36 variables showed no significant association: sexual orientation and education level (sociodemographic variables); number of leisure outings, number of close leisure-related relationships, and community engagement (social variables); extraversion and agreeableness (psychological variables); and sleep duration (mental health variable).

Within the sociodemographic category, relationship status was the strongest correlate of PSMU ($r = -0.140$, 95% CI [−0.202, −0.078], $p < .001$). In the social media use behavior category, the strongest associations were observed for having experienced social media-related problems in the previous three months ($r = 0.453$, 95% CI [0.401, 0.501], $p < .001$), SMUF ($r = 0.433$, 95% CI [0.381, 0.483], $p < .001$), and the perceived need to control SMUF ($r = 0.318$, 95% CI [0.260, 0.373], $p < .001$).

In the social category, higher PSMU was associated with lower satisfaction with intimate relationships ($r = -0.208$, 95% CI [−0.267, −0.147], $p < .001$), lower satisfaction with family and friends' relationships ($r = -0.191$, 95% CI [−0.251, −0.129], $p < .001$), and lower perceived social support ($r = -0.178$, 95% CI [−0.239, −0.117], $p < .001$).

Within the psychological category, higher PSMU was most strongly associated with urgency impulsivity ($r = 0.341$, 95% CI [0.284, 0.395], $p < .001$), overall working memory difficulties ($r = 0.326$, 95% CI [0.268, 0.381], $p < .001$), lower conscientiousness ($r = -0.249$, 95% CI [−0.307, −0.189], $p < .001$), and higher neuroticism ($r = 0.205$, 95% CI [0.144, 0.265], $p < .001$).

Within the mental health category, higher PSMU was associated with greater depressive mood ($r = 0.298$, 95% CI [0.240, 0.354], $p < .001$), more frequent sleep difficulties ($r = 0.199$, 95% CI [0.138, 0.259], $p < .001$), and lower self-esteem ($r = -0.185$, 95% CI [−0.245, −0.124], $p < .001$).

Finally, within the academic IV category, the strongest correlations with participants' PSMU scores were observed for smartphone-related interference with university work outside the classroom ($r = 0.347$, 95% CI [0.290, 0.401], $p < .001$) and inside the classroom ($r = 0.245$, 95% CI [0.185, 0.303], $p < .001$).

3.3. Significant predictors of PSMU (RQ3)

The multivariate linear regression model demonstrated good explanatory power ($R^2 = 0.45$; adjusted $R^2 = 0.43$; RMSE = 0.56), indicating that 45% of the variance in PSMU was explained by the predictor variables. All standard assumptions for multiple linear regression were assessed and satisfied (Starbuck, 2023). Residuals were approximately normally distributed, as indicated by the Shapiro–Wilk test ($W = 0.999$, $p = .677$) and visual inspection of the Q–Q plot. Variance inflation factors (VIFs) were all below 2, indicating no evidence of problematic multicollinearity. No influential observations or heteroscedasticity issues requiring corrective procedures were identified. To preserve the conciseness of the main manuscript, the complete regression diagnostic results and plots are reported in the Supplementary Material (Tables S2–S3, Fig. S1).

Table 2 presents the regression coefficients for all predictors. Positive bêta coefficients (β) indicate that higher predictor values are associated with higher PSMU scores, whereas negative coefficients indicate inverse associations.

Experiencing personal, academic, or professional problems due to social media use in the past three months was the strongest predictor of PSMU ($\beta = 0.55$, 95% CI [0.44, 0.65], $p < .001$), followed by the perceived need to reduce social media use over the past year ($\beta = 0.29$, 95% CI [0.21, 0.38], $p < .001$). Sexual orientation was the third strongest predictor ($\beta = -0.26$, 95% CI [−0.44, −0.08], $p < .001$), with homosexual participants exhibiting higher PSMU scores. Social media use frequency for non-essential activities was the fourth strongest predictor ($\beta = 0.17$, 95% CI [0.13, 0.22], $p < .001$), followed by impulsivity (negative urgency; $\beta = 0.16$, 95% CI [0.09, 0.22], $p < .001$). Gender also significantly predicted PSMU ($\beta = -0.14$, 95% CI [−0.25, −0.03], $p < .001$), indicating higher PSMU scores among women. Higher levels of depressive mood were associated with higher PSMU ($\beta = 0.11$, 95% CI [0.02, 0.20], $p = .013$).

Additional predictors with smaller but significant effects included lower satisfaction with family and friends' relationships ($\beta = -0.05$, 95% CI [−0.10, −0.01], $p = .011$), greater attentional working memory difficulties ($\beta = 0.06$, 95% CI [0.00, 0.11], $p = .032$), lower conscientiousness ($\beta = -0.07$, 95% CI [−0.12, −0.02], $p = .003$), and greater smartphone-related interference with university work outside the classroom ($\beta = 0.03$, 95% CI [0.00, 0.05], $p = .007$).

3.4. Moderators and mediators of the SMUF–PSMU relationship (RQ4)

We tested satisfaction with family and friends' relationships, satisfaction with intimate relationships, perceived social support, community engagement, religious engagement, and connection with the future-self as moderators of the relationship between social media use frequency (SMUF) and PSMU. None of these variables significantly moderated this relationship. The interactions moderation statistics were respectively: $\beta = -0.00$, 95% CI (0.02, −0.05), $p = .968$; $\beta = 0.00$, 95% CI (0.01, −0.02), $p = .783$; $\beta = -0.00$, 95% CI (0.01, −0.03), $p = .646$; $\beta = 0.01$, 95% CI (−0.00, 0.04), $p = .197$; $\beta = 0.00$, 95% CI (−0.02, 0.03), $p = .609$); $\beta = 0.00$, 95% CI (−0.01, 0.02), $p = .609$; $\beta = 0.00$, 95% CI (−0.01, 0.02), $p = .484$. Given the results of the power analysis we reported in the method section, the absence of statistically significant interaction effects indicates that any true moderation effects, if present, were likely to be very small.

Mediation analyses revealed that sleep difficulties, depressive mood, and self-esteem partially mediated the association between SMUF and PSMU. Specifically, significant indirect effects were observed for sleep difficulties (effect = 0.018, 95% CI [0.009, 0.030], $p < .001$; mediation = 5.38%), depressive mood (effect = 0.035, 95% CI [0.022, 0.048], $p < .001$; mediation = 10.3%), and self-esteem (effect = 0.016, 95% CI [0.007, 0.027], $p = .004$; mediation = 4.65%). These findings support partial mediation. However, given the cross-sectional design, causal interpretations cannot be inferred.

4. Discussion

The present study investigated the level, correlates, predictors, and underlying mechanisms of problematic social media use (PSMU) among French psychology university students. By integrating sociodemographic, behavioral, social, psychological, mental health, and academic variables, this work aimed to test hypotheses grounded on previous

Table 2
Multivariate relationships between the independent variables and the score of problematic social media use: linear regression analysis' results.

Variable/measures	*B*	95% CI lower bound	95% CI upper bound	SE	*t*	*p*
Sociodemographic variables						
Age	−0.00482	−0.02350	0.01386	0.00952	−0.507	0.613
Gender						
Female–male	−0.13841	−0.24654	−0.03029	0.05510	−2.512	0.012
Other–male	−0.05914	−0.44852	0.33024	0.19841	−0.298	0.766
Sexual orientation						
Bisexual–heterosexual	0.01681	−0.09315	0.12676	0.05603	0.300	0.764
Homosexual–heterosexual	−0.25968	−0.43687	−0.08249	0.09029	−2.876	0.004
Other–heterosexual	−0.05034	−0.27385	0.17317	0.11389	−0.442	0.659
Relationship status	−0.01759	−0.04273	0.00755	0.01281	−1.373	0.170
Education level (undergraduation–master's degree)	0.01149	−0.08128	0.10426	0.04727	0.243	0.808
Income	−0.01102	−0.08424	0.06219	0.03731	−0.295	0.768
Social media use behavior measures						
Smartphone use frequency for non-essential activities (in last 3 months; measured in hours)	0.17408	0.13157	0.21659	0.02166	8.037	<0.001
Having experienced personal, professional, or academic problems due to smartphone use (previous three months)	0.54675	0.44461	0.64889	0.05205	10.505	<0.001
Perceived need to control smartphone use frequency (previous 12 months)	0.29271	0.20751	0.37792	0.04342	6.742	<0.001
Social activities, relationships, and social engagement measures						
Satisfaction (level of) with family members and friends relationship	−0.05642	−0.10009	−0.01275	0.02225	−2.535	0.011
Satisfaction (level of) with intimate relationships	−0.02340	−0.05628	0.00948	0.01675	−1.397	0.163
Perceived (level of) social support from family and friends when in need	0.01358	−0.01190	0.03906	0.01299	1.046	0.296
Religious (level of) engagement	0.01582	−0.00688	0.03853	0.01157	1.368	0.172
WMQ memory total mean score	0.04331	−0.01687	0.10349	0.03067	1.412	0.158
WMQ attention total mean score	0.06066	0.00539	0.11594	0.02817	2.154	0.032
WMQ executive function total mean score	0.02230	−0.03013	0.07474	0.02672	0.835	0.404
UPPS-P positive urgency total score	0.01840	−0.04666	0.08346	0.03315	0.555	0.579
UPPS-P negative urgency total score	0.15783	0.09225	0.22342	0.03342	4.723	<0.001
BFI consciousness total score	−0.07264	−0.12035	−0.02493	0.02431	−2.988	0.003
BFI-10 neuroticism total score	−0.00419	−0.04572	0.03734	0.02116	−0.198	0.843
BFI-10 openness total score	−0.01907	−0.04804	0.00989	0.01476	−1.292	0.197
Connection (level of) with future-self 20 years from the present	0.00891	−0.00527	0.02310	0.00723	1.233	0.218
Mental health measures						
Sleep difficulties (difficulty falling asleep and/or frequent waking up during the night; experienced in last 6 months)	0.00692	−0.03289	0.19787	0.02029	0.341	0.733
SDHS total score	0.11044	0.02301	0.04924	0.04455	2.479	0.013
SISE self-esteem	−0.00296	−0.05515	0.05616	0.02660	−0.111	0.911
Academic measures						
MPIL (outside classroom) total mean score	0.03263	0.00909	0.05616	0.01199	2.720	0.007
MPIL (inside classroom) total mean score	−0.01135	−0.03381	0.01111	0.01144	−0.992	0.321
Grade mean for the last university semester	0.03990	−0.00137	0.08116	0.02103	1.897	0.058

B = bêta coefficient (estimates); CI = confidence interval; SE = standard error; *t* = *t*-value; *p* = *p*-value (significance at $p < .05$).

studies and derived from problematic use of digital technology, self-regulation, and psychosocial vulnerability models. Overall, the findings largely converge with prior international research while offering novel insights into a population that is theoretically expected to be relatively protected due to heightened awareness of addiction mechanisms.

4.1. Level and prevalence of PSMU

In line with RQ1, the mean PSMU score (M = 2.22, SD = 0.76) was below the scale midpoint (2.5; 1–5 scale). Nevertheless, 15% of participants exceeded a score of 3 points ("descriptive" and less restrictive threshold); 5.5% of participants reported a high score (>3.5 points); 5.6% were classified at high risk of problematic social media use; 3.8% experience meaningful functional impairment ("clinical" and more restrictive method of calculating prevalence). The prevalence based on liberal estimates (15%) are consistent with those reported in European university samples, which typically range between 8% and 25% depending on the measurement approach (Casale et al., 2023; Cheng et al., 2021; Shannon et al., 2022). However, a prevalence based on more conservative estimates (3.8%) are substantially below.

Importantly, these findings yield findings that seemingly challenge the assumption that greater theoretical knowledge of addiction mechanisms necessarily translates into lower vulnerability. Despite having courses on mental health, addiction mechanism and behavior, psychology students appear to experience levels of PSMU comparable to those observed in broader student populations, supporting the notion that awareness alone is insufficient to prevent dysregulated social media use (Billieux et al., 2015; Montag et al., 2021).

It is relevant to note, however, that, as discussed in the introduction, comparisons of PSMU prevalence across studies should be interpreted with caution, as estimates are influenced by differences in assessment instruments, scoring methods, cut-off thresholds, and study populations. Consequently, observed differences in prevalence may reflect methodological variability as well as genuine differences between populations.

4.2. Sociodemographic correlates and predictors

Consistent with prior literature, female gender and sexual minority status emerged as significant correlates and predictors of PSMU (Andreassen et al., 2016; Casale et al., 2023; Fatima et al., 2025). Bisexual and homosexual students reported significantly higher PSMU scores than heterosexual students. Although the magnitude of this difference was small (Sawilowsky, 2009), it remains noteworthy from a public health perspective. Even modest differences between population subgroups may have practical significance when they consistently identify groups at relatively greater risk and help inform targeted prevention and intervention efforts. Overall, the observed association is consistent with previous research suggesting that social media may serve as a compensatory or potentially protective space for identity

exploration, social support, and social connection among sexual minority individuals who experience offline marginalization (Kardefelt-Winther, 2014; Valkenburg et al., 2022). However, these potential mechanisms were not directly assessed in the present study and should therefore be regarded as theoretically plausible explanations rather than mechanisms demonstrated by our findings. Similarly, relationship status was negatively associated with PSMU, with single students reporting higher levels of problematic use. Previous research has suggested that online platforms may, in some circumstances, compensate for limited offline intimacy or social connection (Pazdur et al., 2025), an interpretation that is also compatible with compensatory theories of social media use (Kardefelt-Winther, 2014). Nevertheless, the present data do not establish whether compensatory social media use accounts for the observed association with relationship status. Future studies should directly assess these proposed mechanisms to determine whether they explain the associations observed here.

Contrary to expectations, age, education level, and income were not robust predictors in multivariate models. This may be partly explained by the restricted age range and relative homogeneity of the sample, which limited variability in these sociodemographic factors.

4.3. Central role of social media use behaviors

As hypothesized, behavioral indicators of social media use emerged as the strongest predictors of PSMU. Experiencing personal, academic, or professional problems due to social media use was the most powerful predictor, followed by perceived loss of control and frequency of non-essential use. These findings strongly align with contemporary conceptualizations of PSMU as a disorder of self-regulation rather than mere excessive use (Billieux et al., 2015; Elhai et al., 2017), fact that is also in line with Self-Regulation Theory (Baumeister & Heatherton, 1996) and Temporal Self-Regulation Theory (Hall & Fong, 2007) based perspectives of Internet use.

Notably, the perceived need to control social media use was a strong predictor of PSMU, suggesting that insight into problematic behavior does not necessarily translate into behavioral change. This finding echoes previous work showing that monitoring or control intentions may coexist with persistent dysregulation when underlying cognitive and emotional vulnerabilities are present (Fioravanti et al., 2025; ven Deursen et al., 2015).

4.4. Psychological and personality vulnerabilities

The results strongly support hypotheses regarding psychological vulnerability factors. Consistent with findings from previous studies (Andreassen et al., 2016; Billieux et al., 2021; Carbonell-Colomer et al., 2025), higher impulsivity—particularly urgency—was associated with greater PSMU. However, the observed association should be interpreted with caution, as impulsivity is conceptually and empirically closely related to addictive behaviors. Impulsivity is considered both a vulnerability factor for the development of problematic use of digital technology or addictive behaviors and a characteristic associated with their expression and maintenance. Consequently, the cross-sectional design of the present study does not allow conclusions regarding the directionality or causal nature of this association.

Working memory difficulties, higher neuroticism, and lower conscientiousness were all associated with greater PSMU, consistent with prior research across behavioral addictions (Billieux et al., 2021; Casale & Banchi, 2020; Đurić et al., 2024).

Working memory difficulties were not only linked to PSMU but also to poorer academic performance, reinforcing the role of self-reported executive-function difficulties in attentional capture, habitual checking, and multitasking behaviors (Wilmer et al., 2017). These findings align with dual-process models suggesting that impaired top-down control facilitates compulsive engagement with rewarding digital stimuli (Montag et al., 2021). In addition, the observed association between self-reported executive-function difficulties and PSMU is consistent with theoretical models proposing that impairments in inhibitory control, attention regulation, and goal-directed behavior increase vulnerability to excessive social media use (Hall & Fong, 2007; Kardefelt-Winther, 2014). Conversely, persistent and excessive engagement with social media may further compromise executive functioning, highlighting the likely bidirectional nature of this relationship.

Contrary to expectations, connection with the future-self did not emerge as a significant moderator or predictor in the multivariate model. It is possible that this finding reflects the limited sensitivity of the single-item measure rather than a genuinely weaker role of future-oriented thinking in PSMU compared with other forms of addictive behavior (Sokol & Serper, 2020). Therefore, the present findings should not be interpreted as evidence that future-self connection is unrelated to or plays only a limited role in PSMU. Future studies should further investigate this association using validated multi-item measures of future-self connection and, ideally, longitudinal designs to clarify its potential role in PSMU.

4.5. Mental health and academic functioning

As hypothesized, depressive mood, sleep difficulties, and low self-esteem were significantly associated with higher PSMU. Mediation analyses further demonstrated that these variables partially mediated the relationship between social media use frequency and PSMU, supporting theoretical models that conceptualize PSMU as a maladaptive coping strategy for emotional distress (Bányai et al., 2017; Brand et al., 2025; Vogel et al., 2014). However, the association between depressive mood and PSMU should be interpreted cautiously, as the cross-sectional design precludes conclusions regarding causality. Reverse causality is plausible, whereby depressive symptoms may increase reliance on social media as a coping strategy, while excessive or problematic social media use may, in turn, exacerbate depressive mood. Longitudinal studies are needed to disentangle these potentially bidirectional relationships.

The observed negative association between PSMU and academic performance is consistent with previous findings suggesting that perceived interference and attentional fragmentation, rather than grades per se, constitute the primary academic cost of PSMU (Junco, 2012; Ma, 2025; Pang et al., 2025; Rosen, 2017). However, as the effect-size was small, this finding should be interpreted with caution and should not be taken to imply a substantial impact on academic achievement at the individual level. Further longitudinal research is needed to clarify its practical significance. Also, smartphone-related interference with academic work—especially outside the classroom—was a significant predictor of PSMU, underscoring the reciprocal relationship between academic self-regulation and problematic digital media use.

4.6. Moderation and mediation processes

Contrary to hypotheses, none of the social variables moderated the relationship between SMUF and PSMU. While social resources were negatively correlated with PSMU, they did not buffer the impact of frequent use on problematic outcomes. This might suggest that social connectedness may function as a general protective factor but is insufficient to counteract the effects of intensive use once dysregulation is established.

In contrast, mental health variables partially mediated the SMUF–PSMU relationship, highlighting the central role of emotional and psychological processes. These findings are consistent with bidirectional and transactional models of PSMU, in which excessive use exacerbates distress, which in turn reinforces maladaptive use patterns (Alimoradi et al., 2019; Bányai et al., 2017). However, although the indirect effects were statistically significant, the proportion of the association explained by the mediators was modest. These findings should therefore be interpreted cautiously, suggesting that the mediators

account for only a limited part of the relationship and that additional psychological and contextual mechanisms are likely to be involved.

The present findings converge to reinforce the explanatory value of the Interaction of Person–Affect–Cognition–Execution (I-PACE) model (Brand et al., 2025, 2019, 2016). Rather than identifying a single dominant determinant, the results indicate that PSMU is best understood as the product of dynamic interactions among individual vulnerabilities (e.g., gender, personality traits, impulsivity), affective and cognitive characteristics (e.g., depressive symptoms, self-esteem, executive functioning), and reinforcing engagement with social media, which collectively erode self-regulatory capacity. This integrative pattern lends further support to the I-PACE model as a robust theoretical framework for understanding the etiology and maintenance of PSMU.

5. Limitations and future studies

Several limitations should be acknowledged.

First, the fact the BSMAS psychometric properties have not been specifically established among French university students represents a potential measurement limitation, as the scale's reliability and validity may not fully generalize to this specific population. Consequently, the present findings should be interpreted with appropriate caution, and future studies should specifically evaluate the psychometric properties and measurement validity of the French BSMAS among university students.

Second, the cross-sectional design precludes causal inference, particularly regarding mediation pathways. Longitudinal and experimental designs are needed to disentangle directionality.

Third, all measures relied on self-report, which may be subject to recall bias and social desirability effects, especially in a psychology student sample.

Fourth, several constructs were assessed using single-item measures (e.g., sleep difficulties, self-esteem, relationship satisfaction, perceived social support, and community engagement). Although single-item indicators reduce participant burden and are appropriate for some global constructs, they preclude the assessment of internal consistency and may have introduced measurement error, potentially attenuating the observed associations. Future studies should employ validated multi-item scales to improve measurement reliability. It is relevant to note, however, that the validity of some of the single-item (sleep difficulties, self-esteem; Robins et al., 2001; Vézina-Im et al., 2022) was successfully tested in previous studies through comparisons with close constructs measure by multi-item scales.

Fifth, although agreeableness ($\alpha = 0.23$) and openness ($\alpha = 0.27$) showed low internal consistency, these dimensions were retained because for two reasons: (a) they were measured with the BFI-10, an ultra-short instrument in which each trait is assessed using only two items (Rammstedt & John, 2007) — Cronbach's α is known to underestimate reliability for two-item scales and should be interpreted cautiously (Eisinga et al., 2013); (b) the correlation between the two items of the these dimensions were positive and highly significant. Nevertheless, the low reliability may have attenuated the observed associations; therefore, findings involving these traits should be interpreted cautiously and replicated using longer personality inventories with stronger psychometric properties.

Sixth, because of the anonymity requirements from the ethical committee who approved the study, academic performance was measured using ordinal categories rather than actual grade values. This may reduce analytical precision.

Seventh, the sample was predominantly female and young, which may restrict generalizability to other student populations or age groups. Additionally, although the survey link was made available to all registered students, including those who were absent during in-class recruitment, participation was voluntary. Therefore, the possibility of selection and nonresponse bias cannot be entirely excluded, as students who chose to participate may have differed systematically from those who did not. Such differences may have influenced the observed prevalence and correlates of PSMU and should be considered when interpreting the findings.

Eighth, although the regression model explained 45% of the variance in PSMU, the possibility of some degree of overfitting cannot be completely excluded given the number of predictors (28) included in the multivariate regression model. However, the relatively large sample size (973 participants; approximately 35 observations per predictor) and the absence of problematic multicollinearity support the stability of the model. Future studies should validate these findings in independent samples.

Ninth, problematic social media use may partly reflect broader emotional distress rather than a distinct behavioral condition. Future longitudinal and clinical studies are needed to clarify whether PSMU constitutes an independent condition or primarily represents a maladaptive coping response to underlying psychological difficulties.

6. Conclusion

Despite their presumed awareness of addiction mechanisms due to their academic training, psychology students exhibited levels of problematic social media use seemingly comparable to those reported in similarly aged general population samples. The findings indicate that PSMU was more strongly associated with deficits in self-regulation, emotional vulnerability, cognitive control, and behavioral indicators of impaired self-control than with knowledge about addictive behaviors. In contrast, social resources showed relatively limited associations with PSMU.

Although the present findings cannot establish causal relationships or intervention efficacy, they suggest that preventive strategies may benefit from considering self-regulation, emotional coping, and executive functioning as potential targets for future intervention development and evaluation. Longitudinal and experimental studies are needed to determine the causal mechanisms underlying PSMU and to establish whether interventions targeting these factors effectively reduce problematic social media use.

Abbreviations

DSM-5-TR	Diagnostic and Statistical Manual of Mental Disorders Fifth Edition Text Revision
APA	American Psychiatric Association
ICD-11	International Classification of Diseases
WHO	World Health Organization
PMSU	problematic social media use
SMUF	social media use frequency
RQ	research question
BSMAS	6 item Bergen Social Media Addiction Scale
WMQ	The Working Memory Questionnaire
BFI	The Big Five Inventory
SDHS	Short Depression–Happiness Scale
SISE	The Single-Item Self-Esteem Scale (SISE;
MPIL	Mobile Phone Interference in Life questionnaire
WMA	World Medical Association
IV	independent variables
DV	dependent variable
VIF	variance inflation factors
PV	predictor variables
OV	outcome variable

CRediT authorship contribution statement

Germano Vera Cruz: Writing – original draft, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Clarice Da Rosa:** Methodology, Investigation, Formal analysis, Data curation. **Ilona Lavallée:** Writing – review &

editing, Methodology, Investigation, Data curation. **Eleonor Gilles-Noguès:** Writing – review & editing, Validation, Methodology, Investigation, Data curation. **Yasser Khazaal:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Conceptualization.

Consent for publication

No applicable.

Ethics approval and consent to participate

Ethical approval for this study was conducted by the Ethical Committee of the University of Picardie Jules Verne (Avis n°2026-95/Dossier 2026-01.01). The study was conducted in accordance with the ethical principles outlined in the Declaration of Helsinki. All participants were informed about the purpose of the study, assured of confidentiality, and provided digital informed consent prior to participation. Participation was voluntary, restricted to those aged ≥18 years, and respondents could withdraw at any time without consequence.

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Declaration of competing interest

The authors do not have any conflicts of interest to report.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.actpsy.2026.107797.

Data availability

Data will be made available on request.

References

Aboujaoude, E., Vera Cruz, G., Rochat, L., Courtois, R., Ben Brahim, F., Khan, R., & Khazaal, Y. (2022). Assessment of the popularity and perceived effectiveness of smartphone tools that track and limit smartphone use: Survey study and machine learning analysis. *Journal of Medical Internet Research, 24*(10), Article e38963. https://doi.org/10.2196/38963

Ahmed, O., Walsh, E. I., Dawel, A., Alateeq, K., Espinoza Oyarce, D. A., & Cherbuin, N. (2024). Social media use, mental health and sleep: A systematic review with meta-analyses. *Journal of Affective Disorders, 367*, 701–712. https://doi.org/10.1016/j.jad.2024.08.193. Dec 15.

Alimoradi, Z., Lin, C. Y., Broström, A., Bülow, P. H., Bajalan, Z., Griffiths, M. D., … Pakpour, A. H. (2019). Internet addiction and sleep problems: A systematic review and meta-analysis. *Sleep Medicine Reviews, 47*, 51–61. https://doi.org/10.1016/j.smrv.2019.06.004

Alwuqaysi, B. (2025). A systematic review and meta-analysis of social media's influence on mental health, family functioning, and visual content. *Social Sciences & Humanities Open., 12*, Article 101793. https://doi.org/10.1016/j.ssaho.2025.101793

American Psychiatric Association (APA). (2022). *Diagnostic and statistical manual of mental disorders* (5th ed.). Washington, DC: American Psychiatric Association. Text Revision.

Andreassen, C. S., Billieux, J., Griffiths, M. D., Kuss, D. J., Demetrovics, Z., Mazzoni, E., et al. (2016). The relationship between addictive use of social media and video games and symptoms of psychiatric disorders: A large-scale cross-sectional study. *Psychology of Addictive Behaviors, 30*(2), 252–262. https://doi.org/10.1037/adb0000160

Andreassen, C. S., & Pallesen, S. (2014). Social network site addiction — An overview. *Current Pharmaceutical Design, 20*(25), 4053–4061. https://doi.org/10.2174/13816128113199990616

Arigo, D., Bercovitz, I., Lapitan, E., et al. (2024). Social comparison and mental health. *Current Treatment Options in Psychiatry, 11*, 17–33. https://doi.org/10.1007/s40501-024-00313-0

Babyak, M. A. (2004). What you see may not be what you get: A brief, nontechnical introduction to overfitting in regression-type models. *Psychosomatic Medicine, 66*(3), 411–421. https://doi.org/10.1097/01.psy.0000127692.23278.a9

Bányai, F., Zsila, Á., Király, O., Maraz, A., Elekes, Z., Griffiths, M. D., … Demetrovics, Z. (2017). Problematic social media use: Results from a large-scale nationally representative adolescent sample. *PLoS One, 12*(1), Article e0169839. https://doi.org/10.1371/journal.pone.0169839

Baumeister, R. F., & Heatherton, T. F. (1996). Self-regulation failure: An overview. *Psychological Inquiry, 7*(1), 1–15. https://doi.org/10.1207/s15327965pli0701_1

Beiter, R., Nash, R., McCrady, M., Rhoades, D., Linscomb, M., Clarahan, M., & Sammut, S. (2015). The prevalence and correlates of depression, anxiety, and stress in a sample of college students. *Journal of Affective Disorders, 173*, 90–96. https://doi.org/10.1016/j.jad.2014.10.054

Billieux, J., Heeren, A., Rochat, L., Maurage, P., Bayard, S., Bet, R., … Baggio, S. (2021). Positive and negative urgency as a single coherent construct: Evidence from a large-scale network analysis in clinical and non-clinical samples. *Journal of Personality, 89*(6), 1252–1262. https://doi.org/10.1111/jopy.12655

Billieux, J., Maurage, P., Lopez-Fernandez, O., Kuss, D. J., & Griffiths, M. D. (2015). Can disordered mobile phone use be considered a behavioral addiction? An update on current evidence and a comprehensive model for future research. *Current Addiction Reports, 2*, 156–162. https://doi.org/10.1007/s40429-015-0054-y

Billieux, J., Rochat, L., Ceschi, G., Carré, A., Offerlin-Meyer, I., Defeldre, A. C., … Van der Linden, M. (2012). Validation of a short French version of the UPPS-P Impulsive Behavior Scale. *Comprehensive Psychiatry, 53*(5), 609–615. https://doi.org/10.1016/j.comppsych.2011.09.001

Boer, M., Stevens, G. W. J. M., Finkenauer, C., de Looze, M. E., & van den Eijnden, R. J. J. M. (2021). Social media use intensity, social media use problems, and mental health among adolescents: Investigating directionality and mediating processes. *Computers in Human Behavior, 116*, Article 106645. https://doi.org/10.1016/j.chb.2020.106645

Bowman, L. L., Levine, L. E., Waite, B. M., & Gendron, M. (2010). Can students really multitask? An experimental study of instant messaging while reading. *Computers in Education, 54*(4), 927–931. https://doi.org/10.1016/j.compedu.2009.09.024

Brailovskaia, J., & Margraf, J. (2024). Addictive social media use during Covid-19 outbreak: Validation of the Bergen Social Media Addiction Scale (BSMAS) and investigation of protective factors in nine countries. *Current Psychology, 43*, 13022–13040. https://doi.org/10.1007/s12144-022-03182-z

Brand, M., Müller, A., Wegmann, E., Antons, S., Brandtner, A., Müller, S. M., … Potenza, M. N. (2025). Current interpretations of the I-PACE model of behavioral addictions. *Journal of Behavioral Addictions, 14*(1), 1–17. https://doi.org/10.1556/2006.2025.00020

Brand, M., Wegmann, E., Stark, R., Müller, A., Wölfling, K., Robbins, T. W., & Potenza, M. N. (2019). The Interaction of Person-Affect-Cognition-Execution (I-PACE) model for addictive behaviors: Update, generalization to addictive behaviors beyond internet-use disorders, and specification of the process character of addictive behaviors. *Neuroscience and Biobehavioral Reviews, 104*, 1–10. https://doi.org/10.1016/j.neubiorev.2019.06.032

Brand, M., Young, K. S., Laier, C., Wölfling, K., & Potenza, M. N. (2016). Integrating psychological and neurobiological considerations regarding the development and maintenance of specific Internet-use disorders: An Interaction of Person-Affect-Cognition-Execution (I-PACE) model. *Neuroscience and Biobehavioral Reviews, 71*, 252–266. https://doi.org/10.1016/j.neubiorev.2016.08.033

Carbonell-Colomer, M., Marchena-Giráldez, C., & Bernabéu-Brotóns, E. (2025). Social media and functional deterioration: Indicators of problematic use in university students. *Frontiers in Psychology, 16*, Article 1720760. https://doi.org/10.3389/fpsyg.2025.1720760

Carver, C. S., & Scheier, M. F. (1998). *On the self-regulation of behavior*. Cambridge University Press. https://doi.org/10.1017/CBO9781139174794

Casale, S., Akbari, M., Seydavi, M., Benucci, S. B., & Fioravanti, G. (2023). Has the prevalence of problematic social media use increased over the past seven years and since the start of the COVID-19 pandemic? A meta-analysis of the studies published since the development of the Bergen social media addiction scale. *Addictive Behaviors, 147*, Article 107838. https://doi.org/10.1016/j.addbeh.2023.107838

Casale, S., & Banchi, V. (2020). Narcissism and problematic social media use: A systematic literature review. *Addictive Behaviors Reports, 11*, Article 100252. https://doi.org/10.1016/j.abrep.2020.100252

Cheng, C., Lau, Y. C., Chan, L., & Luk, J. W. (2021). Prevalence of social media addiction across 32 nations: Meta-analysis with subgroup analysis of classification schemes and cultural values. *Addictive Behaviors, 117*, Article 106845. https://doi.org/10.1016/j.addbeh.2021.106845

Corke, L., Maksyutynska, K., Jones, J. M., & George, T. P. (2025). The relationship between social media use, and mental health disorders in adolescents and young adults: A scoping review. *Annals of Clinical Psychiatry, 36*(3), 89–99. https://doi.org/10.1177/10401237251344098

David, P., Kim, J. H., Brickman, J. S., Ran, W., & Curtis, C. M. (2015). Mobile phone distraction while studying. *New Media & Society, 17*(10), 1661–1679. https://doi.org/10.1177/1461444814531692

Dormann, C. F., Elith, J., Bacher, S., Buchmann, C., Carl, G., Carré, G., … Lautenbach, S. (2013). Collinearity: A review of methods to deal with it and a simulation study evaluating their performance. *Ecography, 36*, 27–46. https://doi.org/10.1111/j.1600-0587.2012.07348.x

Đurić, P., Ćudina, V., Bolić, T., & Glavak-Tkalić, R. (2024). Personality traits, self-esteem, life satisfaction and problematic social media use among young adults. *Journal of European Psychology Students, 14*(2), 12–26. https://doi.org/10.59477/jeps.598

Efron, B., & Tibshirani, R. J. (1993). *An introduction to the bootstrap*. Chapman & Hall.

Eisinga, R., Grotenhuis, M., & Pelzer, B. (2013). The reliability of a two-item scale: Pearson, Cronbach, or Spearman-Brown? *International Journal of Public Health, 58*(4), 637–642. https://doi.org/10.1007/s00038-012-0416-3. Aug.

Elhai, J. D., Dvorak, R. D., Levine, J. C., & Hall, B. J. (2017). Problematic smartphone use: A conceptual overview and systematic review of relations with anxiety and depression psychopathology. *Journal of Affective Disorders, 207*, 251–259. https://doi.org/10.1016/j.jad.2016.08.030. Jan 1.

Fatima, A., Akhter, M. S., Kanekar, A., Roy, S., Mitra, R., Imade, B., & Sharma, M. (2025). A scoping review of the use and determinants of social media among college students. *Healthcare (Basel), 13*(17), Article 2234. https://doi.org/10.3390/healthcare13172234

Fioravanti, G., Benucci, S. B., & Ghinassi, S. (2025). Psychological risk factors for problematic social network use: An overview of systematic reviews and meta-analyses. *Addictive Behaviors Reports, 21*, Article 100600. https://doi.org/10.1016/j.abrep.2025.100600

Fioravanti, G., Casale, S., Bocci Benucci, S. B., Prostamo, A., Falone, A., Ricca, V., & Rotella, F. (2021). Fear of missing out and social networking sites use and abuse: A meta-analysis. *Computers in Human Behavior, 122*, Article 106839. https://doi.org/10.1016/j.chb.2021.106839

Griffiths, M. (2005). A 'components' model of addiction within a biopsychosocial framework. *Journal of Substance Use, 10*(4), 191–197. https://doi.org/10.1080/14659890500114359

Hall, P. A., & Fong, G. T. (2007). Temporal self-regulation theory: A model for individual health behavior. *Health Psychology Review, 1*(1), 6–52. https://doi.org/10.1080/17437190701492437

Harrell, F. E., Jr. (2015). *Regression modeling strategies: With applications to linear models, logistic and ordinal regression, and survival analysis* (2nd ed.). Springer. https://doi.org/10.1007/978-3-319-19425-7

Hayes, A. F. (2018). *Introduction to mediation, moderation, and conditional process analysis: A regression-based approach (methodology in the social sciences)* (2nd ed.). New York, NY: The Guilford Press.

Hershfield, H. (2023). *Your future self: How to make tomorrow better today*. Little Brwon.

Ibrahim, A. K., Kelly, S. J., Adams, C. E., & Glazebrook, C. (2013). A systematic review of studies of depression prevalence in university students. *Journal of Psychiatric Research, 47*(3), 391–400. https://doi.org/10.1016/j.jpsychires.2012.11.015

Joseph, S., Linley, P. A., Harwood, J., Lewis, C. A., & McCollam, P. (2004). Rapid assessment of well-being: The Short Depression-Happiness Scale (SDHS). *Psychology and Psychotherapy: Theory, Research and Practice, 77*(4), 463–478. https://doi.org/10.1348/1476083042555406

Junco, R. (2012). In-class multitasking and academic performance. *Computers in Human Behavior, 28*(6), 2236–2243. https://doi.org/10.1016/j.chb.2012.06.031

Kardefelt-Winther, D. (2014). A conceptual and methodological critique of Internet addiction research: Towards a model of compensatory Internet use. *Computers in Human Behavior, 31*, 351–354. https://doi.org/10.1016/j.chb.2013.10.059

Karim, F., Oyewande, A. A., Abdalla, L. F., Chaudhry Ehsanullah, R., & Khan, S. (2020). Social media use and its connection to mental health: A systematic review. *Cureus, 12*(6), e8627. https://doi.org/10.7759/cureus.8627. Jun 15.

Katz, E., Blumler, J. G., & Gurevitch, M. (1973–1974). Uses and gratifications research. *The Public Opinion Quarterly, 37*(4), 509–523. https://doi.org/10.1086/268109

Kircaburun, K., Alhabash, S., Tosuntaş, Ş. B., et al. (2020). Uses and gratifications of problematic social media use among university students: A simultaneous examination of the Big Five of personality traits, social media platforms, and social media use motives. *International Journal of Mental Health and Addiction, 18*, 525–547. https://doi.org/10.1007/s11469-018-9940-6

Kuss, D. J., & Griffiths, M. D. (2017). Social networking sites and addiction: Ten lessons learned. *International Journal of Environmental Research and Public Health, 14*(3), Article 311. https://doi.org/10.3390/ijerph14030311

Lepp, A., Barkley, J. E., & Karpinski, A. C. (2015). The relationship between cell phone use and academic performance in a sample of U.S. college students. *SAGE Open, 5*(1). https://doi.org/10.1177/2158244015573169

Ma, X. (2025). The relationship between psychological stress and academic performance among college students: The mediating roles of cognitive load and self-efficacy. *Acta Psychologica, 259*, Article 105433. https://doi.org/10.1016/j.actpsy.2025.105433

Marino, C. (2018). Quality of social-media use may matter more than frequency of use for adolescents' depression. *Clinical Psychological Science, 6*(4), 455. https://doi.org/10.1177/2167702618771979

McKenna, K. Y. A., & Bargh, J. A. (2000). Plan 9 from cyberspace: The implications of the internet for personality and social psychology. *Personality and Social Psychology Review, 4*(1), 57–75. https://doi.org/10.1207/S15327957PSPR0401_6

Ministère de l''Enseignement Supérieurs, de la Recherche et de l''Espace. (2026). Les effectifs étudiants dans l'enseignement supérieur en 2025–2026. Available from: https://data.esr.gouv.fr/publications/zenodo-19680139.

Montag, C., Haibo Yang, H., & Elhai, J. D. (2021). On the psychology of TikTok use: A first glimpse from empirical findings. *Frontiers in Public Health, 9*. https://doi.org/10.3389/fpubh.2021.641673

Moretta, T., & Wegmann, E. (2025). Toward the classification of social media use disorder: Clinical characterization and proposed diagnostic criteria. *Addictive Behaviors Reports, 21*, Article 100603. https://doi.org/10.1016/j.abrep.2025.100603

Observatoire des Inégalités. (2025). Les femmes devancent les hommes dans l'enseignement supérieur, mais pas dans les filières les plus valorisées [Women outnumber men in higher education, but not in the most prestigious fields of study]. Available from: https://www.inegalites.fr/etudes-superieures-selon-le-sexe.

Pang, H., Jin, X., & Zhang, W. (2025). Deciphering the underlying repercussions of cognitive overload on university students' fatigue, frustration and academic productivity: Implementation of stimulus–organism–response model. *Acta Psychologica, 260*, Article 105469. https://doi.org/10.1016/j.actpsy.2025.105469

Pazdur, M., Tutus, D., & Haag, A. C. (2025). Risk factors for problematic social media use in youth: A systematic review of longitudinal studies. *Adolescent Research Review, 10*, 237–253. https://doi.org/10.1007/s40894-025-00264-4

Peng, P., Chen, Z., Ren, S., et al. (2023). Determination of the cutoff point for smartphone application-based addiction scale for adolescents: A latent profile analysis. *BMC Psychiatry, 23*, 675. https://doi.org/10.1186/s12888-023-05170-4

Preacher, K. J., & Hayes, A. F. (2008). Asymptotic and resampling strategies for assessing and comparing indirect effects in multiple mediator models. *Behavior Research Methods, 40*, 879–891. https://doi.org/10.3758/BRM.40.3.879

Rammstedt, B., & John, O. P. (2007). Measuring personality in one minute or less: A 10 item short version of the Big Five Inventory in English and German. *Journal of Research in Personality, 41*, 203–212.

Robins, R. W., Hendin, H. M., & Trzesniewski, K. H. (2001). Measuring global self-esteem: Construct validation of a single-item measure and the Rosenberg Self-Esteem Scale. *Personality and Social Psychology Bulletin, 27*(2), 151–161. https://doi.org/10.1177/01461672012720

Rochat, L., Vera Cruz, G., Aboujaoude, E., Courtois, R., Brahim, F. B., Khan, R., & Khazaal, Y. (2025). Problematic smartphone use in a representative sample of US adults: Prevalence and predictors. *Addictive Behaviors, 162*, Article 108228. https://doi.org/10.1016/j.addbeh.2024.108228

Rosen, L. D. (2017). The distracted student mind — Enhancing its focus and attention. *Phi Delta Kappan, 99*(2), 8–14. https://doi.org/10.1177/0031721717734183

Rousseau, A., & Rodgers, R. F. (2025). Social media incidental appearance exposure and young people's body image: A conceptual review. *Body Image, 52*, Article 101838. https://doi.org/10.1016/j.bodyim.2024.101838

Rubin, A. M. (2009). Uses-and-gratifications perspective on media effects. In J. Bryant, & M. B. Oliver (Eds.), *Media effects* (pp. 181–200). New York: Routledge.

Sawilowsky, S. S. (2009). New effect size rules of thumb. *Journal of Modern Applied Statistical Methods, 8*(2), 597–599. https://doi.org/10.22237/jmasm/1257035100

Shannon, H., Bush, K., Villeneuve, P. J., Hellemans, K. G., & Guimond, S. (2022). Problematic social media use in adolescents and young adults: Systematic review and meta-analysis. *JMIR Mental Health, 9*(4), Article e33450. https://doi.org/10.2196/33450

Sokol, Y., & Serper, M. (2020). Development and validation of a future self-continuity questionnaire: A preliminary report. *Journal of Personality Assessment, 102*(5), 677–688. https://doi.org/10.1080/00223891.2019.1611588

Starbuck, C. (2023). Linear regression. In *The fundamentals of people analytics*. Cham: Springer. https://doi.org/10.1007/978-3-031-28674-2_10.

Valkenburg, P. M., Meier, A., & Beyens, I. (2022). Social media use and its impact on adolescent mental health: An umbrella review of the evidence. *Current Opinion in Psychology, 44*, 58–68. https://doi.org/10.1016/j.copsyc.2021.08.017

Valkenburg, P. M., & Peter, J. (2027). Online communication and adolescent well-being: Testing the stimulation versus the displacement hypothesis. *Journal of Computer-Mediated Communication, 12*(4), 1169–1182. https://doi.org/10.1111/j.1083-6101.2007.00368.x

Vallat-Azouvi, C., Pradat-Diehl, P., & Azouvi, P. (2012). The Working Memory Questionnaire: A scale to assess everyday life problems related to deficits of working memory in brain injured patients. *Neuropsychological Rehabilitation, 22*(4), 634–649. https://doi.org/10.1080/09602011.2012.681110

ven Deursen, A., Bolle, C. L., Hegner, S., & Kommers, P. (2015). Modeling habitual and addictive smartphone behavior: The role of smartphone usage types, emotional intelligence, social stress, self-regulation, age, and gender. *Computers in Human Behavior, 45*, 411–420. https://doi.org/10.1016/j.chb.2014.12.039

Vera Cruz, G., Aboujaoude, E., Khan, R., Rochat, R., Ben Brahim, F., Courtois, R., & Khazaal, Y. (2023). Smartphone apps for mental health and wellbeing: A usage survey and machine learning analysis of psychological and behavioral predictors. *Digital Health, 9*. https://doi.org/10.1177/20552076231152164

Vera Cruz, G., & Da Rosa, C. (2026). Distinct profiles of problematic smartphone and social media use among French university students: A person-centered analysis of sociodemographic, psychological, social, and academic correlates. *Psychiatry International., 7*(4), Article 185. https://doi.org/10.3390/psychiatryint7040185

Vera Cruz, G., Da Rosa, C., & Khazaal, K. (2026). Demographic, social, psychological, mental health, and academic correlates of problematic smartphone use among French psychology students. *Frontiers in Digital Health, 8*, Article 1784927. https://doi.org/10.3389/fdgth.2026.1784927

Vézina-Im, L.-A., Morin, C. M., Turcotte, S., & Desroches, S. (2022). Validation de la version française du Sleep Health Index: L'index Sur la santé du sommeil [Validation of the French version of the Sleep Health Index: The sleep health index]. *Médecine Du Sommeil., 19*(2), 78–88. https://doi.org/10.1016/j.msom.2021.11.003

Vogel, E. A., Rose, J. P., Roberts, L. R., & Eckles, K. (2014). Social comparison, social media, and self-esteem. *Psychology of Popular Media Culture, 3*(4), 206–222. https://doi.org/10.1037/ppm0000047

Wilmer, H. H., Sherman, L. E., & Chein, J. M. (2017). Smartphones and cognition: A review of research exploring the links between mobile technology habits and cognitive functioning. *Frontiers in Psychology, 8*, Article 605. https://doi.org/10.3389/fpsyg.2017.00605

World Health Organization (WHO). (2022). *International Classification of Diseases (ICD-11)*. Geneva: WHO. Available from: https://www.who.int/standards/classifications/classification-of-diseases (accessed March 18, 2026).

World Medical Association (WMA). (2024). *WMA declaration of Helsinki — Ethical principles for medical research involving human participants*. WMA. Available from: htt

ps://www.wma.net/policies-post/wma-declaration-of-helsinki/ (accessed March 18, 2026)..

Further Reading

Shapiro, S. S., & Wilk, M. B. (1965). An analysis of variance test for normality (complete samples). *Biometrika, 52*(3–4), 591–611. https://doi.org/10.1093/biomet/52.3-4.591